Name: _____

Date: _____

Mr. Rawson • Y9 MYP Physics

2.7 – Design Challenge — Light the Display

Unit 2: Electricity & Energy

Learning intentions

The brief: the school's winter display needs **3 identical LEDs**. Each LED needs exactly **2 V** and **0.3 A** to glow safely — too much of either and it burns out. The only power supply available is a **6 V** battery pack. **Design the circuit.**

No new ideas today — everything you need is from the last six lessons. This is where you put it all together, on your own.

Step 1 — Choose your arrangement

1) **Decide** series or parallel for your 3 LEDs, and justify your choice using **one** fact from Lesson 3 or 4 about what happens if **one** LED fails.

Model answer: *Either answer can be justified. Series: if one LED fails, the loop breaks and all three go out (the current rule means there is only one path). Parallel: if one LED fails, the other two branches are unaffected, because each branch still gets the full supply voltage regardless of the others.*

Step 2 — Check the voltage

2) **Calculate** whether your chosen arrangement delivers exactly 2 V to each LED straight from the 6 V supply, or whether a resistor is needed to drop the spare voltage. If a resistor is needed, calculate its value, using 0.3 A as the current through it.

Model answer: Series: $3 \times 2\,V = 6\,V$ *— exactly the supply voltage (voltage rule, L3). No resistor needed.* ***Parallel:*** *each branch sees the full 6 V (voltage rule, L4), but the LED only wants 2 V, so a resistor must drop the spare* $6 - 2 = 4\,V$ *at 0.3 A:* $R = \dfrac{V}{I} = \dfrac{4}{0.3} \approx 13.3\,\Omega$ *in series with* ***each*** *LED.*

Step 3 — Draw it

3) **Draw** the full circuit diagram for your design: the battery, a switch, all three LEDs, and any resistors your Step 2 answer needs. Label every symbol.

Series design: one loop — battery, switch, three LED symbols, no resistors. Parallel design: battery and switch on the main leg, three branches off two shared bars, each branch containing one LED ***and*** *one* $13.3\,\Omega$ *resistor in series within that branch.*

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Step 4 — Check it

4) **Draw** an ammeter and a voltmeter onto your Step 3 diagram so that, together, they **prove** one LED is getting exactly 2 V and 0.3 A — not just that the whole display lights up.

> *An ammeter in series with (immediately before or after) one LED — and, for the parallel design, before its resistor rather than measuring the whole branch's components separately is also acceptable, since current is the same throughout one branch. A voltmeter connected across that same LED only, not across the LED-and-resistor pair.*

Step 5 — Power

5) **Calculate** the **total** power the battery supplies to your design.

Model answer: Series: *current is 0.3 A throughout (current rule), so* $P = V \times I = 6 \times 0.3 = 1.8\,W$. ***Parallel:*** *total current is* $3 \times 0.3 = 0.9\,A$ *(current rule, L4), so* $P = 6 \times 0.9 = 5.4\,W$.

Extension — which design actually wastes energy

Of the power you calculated in Step 5, only the power actually used *by the LEDs* ($3 \times 2\,\text{V} \times 0.3\,\text{A} = 1.8\,\text{W}$) does anything useful — the rest is wasted as heat in any resistors.

6) **State** which design — series or parallel — wastes energy as heat, and how much of its total power (as a fraction or percentage) is wasted.

Model answer: *The series design wastes nothing — all 1.8 W goes to the LEDs, because it needed no resistor. The parallel design wastes* $5.4 - 1.8 = 3.6\,W$ *out of its 5.4 W total — exactly two-thirds (about 67%), lost as heat in the three resistors.*

HOMEWORK

The display is being expanded from 3 LEDs to **60**. Would you keep your Step 1 design, or switch to the other arrangement? Write two or three sentences, weighing up what you found in Steps 1–5 at this much larger scale.